

☒ Detailed Theory: Snort IDS, Architecture & Evasion Techniques

☒ 1. Snort – Overview

Snort is an open-source **Network Intrusion Detection System (NIDS)** developed by Cisco (initially by Martin Roesch). It can operate in 3 modes:

1.

Sniffer Mode – Monitors and displays packets.

2.

Packet Logger Mode – Saves packets to disk.

3.

NIDS Mode – Analyzes traffic using rules to detect threats.

☒ 2. Installing & Testing Snort

☒ Snort on Windows

1.

Download Snort for Windows (e.g., from [Snort.org](https://www.snort.org)).

2.

Install **WinPcap/Npcap** to allow packet capture.

3.

Configure snort.conf:

-

Set paths to rules and logs

-

Configure network variables (HOME_NET, EXTERNAL_NET)

4.

Run in CMD:

```
bash
```

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```
snort -c C:\Snort\etc\snort.conf -i 2 -A console
```

☒ Snort on Linux

1.

Install using APT/YUM:

```
bash
```

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```
sudo apt install snort
```

2.

Set interface and config in /etc/snort/snort.conf

3.

Example run command:

```
bash
```

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```
sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
```

4.

Test with ICMP (ping):

```
bash
```

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```
ping <some_ip>
```

If a relevant ICMP rule exists, Snort will alert.

❏ 3. IDS Architecture

The IDS architecture has three key layers:

Layer	Description
Data Collection (Sensors)	Captures packets from the network
Analysis (Detection Engine)	Applies rules/patterns to detect threats
Response (Alerting/Logging)	Generates alerts, logs data

Snort fits in this architecture by acting as **sensor + analyzer + alert engine**.

❏ 4. Bypassing an IDS

Attackers may try to evade detection by:

☒ 1. Obfuscation Techniques

- Fragmentation of packets
- Encoding payloads (Unicode, Base64)
- Using uncommon ports
- Altering case in strings

☒ 2. Timing Evasion

- Slow-rate attacks (Low & Slow)
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Time-distributed scans

☒ 3. Payload Obfuscation

- Encrypting payload (SSL)
- Using polymorphic shellcode

☒ 4. Application-Layer Attacks

- Hide payloads in legitimate HTTP/HTTPS requests

☒ 5. Snort Rule Example

snort

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```
alert tcp any any -> any 80 (msg:"Possible Web Exploit"; content:"/cmd.exe"; sid:1000001;)
```

This rule detects HTTP traffic with /cmd.exe, which may indicate an attempted Windows command execution via web.

❏ 6. Snort Logs and Alert Types

- **Fast alert** – Lightweight, readable alerts
 - **Full alert** – Detailed logs with packet dumps
 - **Unified2** – Binary format for third-party tools (e.g., Snorby, BASE)
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☒ 7. Improving Detection

To minimize bypasses:

- Enable **frag3**, **stream5**, and preprocessors in Snort
- Use **SSL decryption** with proxies
- Correlate with logs (SIEM integration)
- Use behavioral detection alongside signature-based